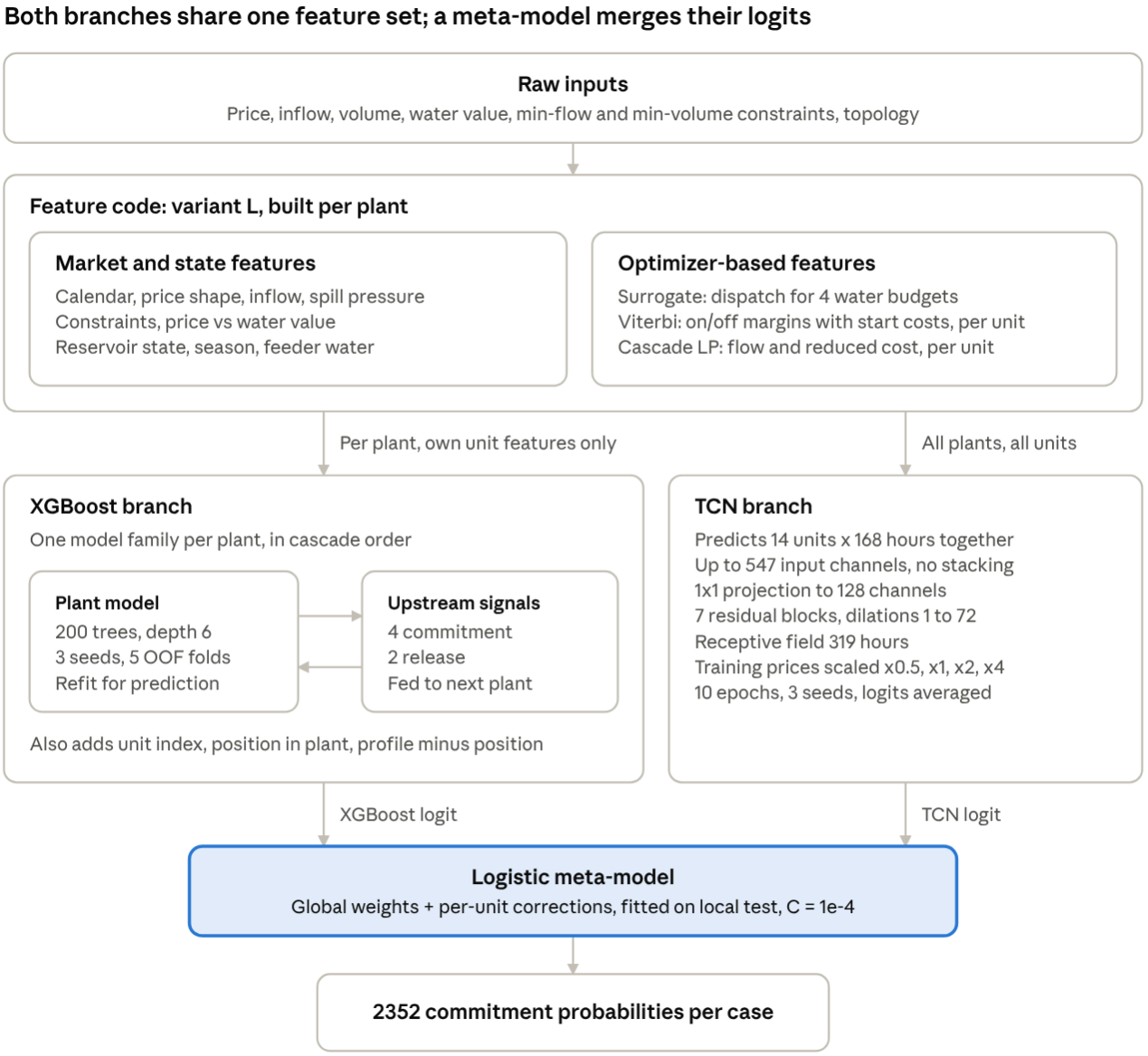
Model v18: XGBoost + TCN with a logistic meta-model

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Overview

v18 predicts whether each of 14 generators is on in each of 168 hours, by combining a physics-informed XGBoost cascade with a temporal convolutional network (TCN). On the local test set the meta-model reaches 0.9862 micro AUC in cross-validation, against 0.9825 for XGBoost and 0.9723 for the TCN. Its recorded Kaggle public score is 0.99161.



v18 architecture: feature flow from raw inputs to the final output

The same feature code feeds both branches. XGBoost sees one plant at a time and receives upstream predictions, while the TCN sees all plants at once and models the week through dilated convolutions.

Data

Each case is one week starting on a given day: 168 hours of inputs and 168 hours of on/off targets for 14 generators, or 2352 binary outputs.

Source	Resolution	Used as
Day-ahead price	Hourly, 168 values	Price features, surrogate, Viterbi, LP
Inflow, 18 series	Hourly, 168 values	Inflow, spill pressure, water budget, LP
Reservoir volume, 17 reservoirs	Value at case start	Head, fill level, capacity, water budget, LP
Synthetic water value, 17 reservoirs	Values at start and end of the week	Price-to-water comparisons, dispatch thresholds, LP
Min-flow and min-volume constraints, 9 series	Hourly, 168 values	Constraint features, surrogate, LP
Topology YAML	Fixed	Plant graph, head curves, losses, turbine curves, start and stop costs

The base table has 3243 columns per case: 19 hourly series × 168, plus 17 × 3 reservoir values. The models never see this table directly; they see the features below.

Run	Train cases	Prediction cases	Case weights
Local validation	First 2337	Last 578, after a 7-case gap	None
Kaggle submission	All 2922	725	Cases from 2021 onward count double

Features

Both branches use the same feature code (`base`, variant L), computed plant by plant. Only the XGBoost branch gets upstream predictions and unit identity columns. Counts are per plant unless stated, and were derived from the code rather than printed by a run.

In the counts, R is the plant's source reservoirs plus its downstream reservoir, when it has one.

Group	Features	Count	XGBoost	TCN
Calendar	Hour of week, hour of day, day of week	3	Yes	Yes
Price shape	log(P / week mean), the same 1 h earlier and 1 h later, rank in week, rank in day, (P − day mean) / sd, (P − week mean) / sd, log(P / final water value of first source)	8	Yes	Yes
Inflow	Hourly inflow summed over source reservoirs, its weekly mean	2	Yes	Yes
Upstream commitment	Mean upstream probability, lagged 1 h, lagged 3 h, weekly mean	4	Yes	No
Constraints	Hourly series listed per plant below, volume above minimum, downstream shortfall; Lio adds required release, its cumulative sum and weekly total	2 to 9	Yes	Yes
Price vs water value	log(P / water value) at start and end of week, for each reservoir in R	2R	Yes	Yes
Spill pressure	Cumulative inflow / capacity, (volume + cumulative inflow − max volume) / capacity	2	Yes	Yes

Group	Features	Count	XGBoost	TCN
Optimizer surrogate	Hourly dispatch profile filling the highest-priced hours, and the water budget as a share of the week, for 4 budgets: minimum to avoid spill, water value at start, water value at end, maximum usable	8	Yes	Yes
Upstream release	Expected upstream release / capacity, and its cumulative sum; the release also raises the surrogate's budgets	2	Yes	No
Reservoir state	Volume, log water value at start and end, and fill level for each reservoir in R; head at first source, net head, log water-value difference between source and downstream	4R + 3	Yes	Yes
Season and feeder water	Day of year, feeder-reservoir volume plus inflow / capacity	2	Yes	Yes
Unit physics and start costs	Viterbi on/off margin for 3 price margins (3), its sign for 2 of them (2), marginal power ratio, log(budget-adjusted threshold / water value)	7 per unit	Own unit	All units
Cascade LP	Relative flow and reduced cost / water value for the unit	2 per unit	Own unit	All units
Unit identity	Unit index, position in the plant, surrogate profile (water value at end) minus position	3	Yes	Profile only, without the subtraction

The three physics-based parts deserve a precise definition:

- **Optimizer surrogate:** the water budget W becomes a schedule through clip(W / capacity − price rank, 0, 1) × number of units. It is a fast approximation of the scheduling decision for one plant.
- **Viterbi margins:** a two-state forward and backward pass gives the best weekly value with the unit on at hour t, minus the best value with it off, after start and stop costs. The margins are P − w, ρP − w and ρP − λ. Here w is the water value, ρ the unit's marginal power ratio and λ the threshold moved by the water budget.
- **Cascade LP:** one weekly linear program over all reservoirs, rivers, tunnels and generators, with penalised slack on the minimum constraints. It is solved once per case and has no binary commitment variables, so its flows and reduced costs are soft signals.

Plant	Units	Sources	Downstream	Upstream plants	Hourly constraint series	XGBoost columns per row	TCN channels
Vesle Kjela	1	Kjelavatn	Foersvatn	None	b_Kjelavatn	55	53
Songa	1	Songavatn, Bitdalsvatn	Totak	None	Totak	61	59
Byrte	1	Botnedalsvatn	Byrtevatn	None	b_Byrtevatn, Byrtevatn	56	54
Haukeli	1	Langeidvatn, Vatjern	Vinjevatn	None	None	60	58
Kjela	1	Foersvatn, Bordalsvatn	Hyljelihyl	Vesle Kjela	b_Hyljelihyl	67	59
Lio	1	Byrtevatn	Bandak	Byrte	r_Vest_Vassdraget, b_Byrtevatn, b_Vinjevatn, Byrtevatn	67	59
Vinje	3	Vaamarvatn, Totak	Vinjevatn	Kjela, Songa	Totak, b_Vinjevatn	68	78
Tokke	4	Vinjevatn	Bandak	Vinje, Haukeli	b_Vinjevatn	61	80
Hogga	1	Bandak	None	Tokke, Lio	b_Bandak	55	47

The TCN channels sum to 547 before constant channels are removed. Vinje's surrogate and unit margins use Totak's water value, and Haukeli's use Langeidvatn's. Haukeli's capacity and head use Vatjern.

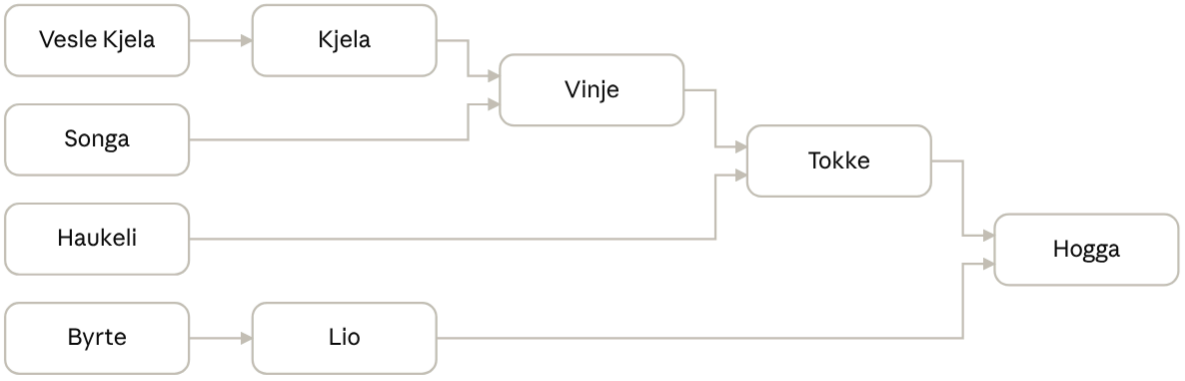
XGBoost branch

The branch trains one model family per plant, in cascade order, so downstream plants can use upstream predictions. Units of a plant are stacked as rows, one row per unit and hour, and each row carries only that unit's physics and LP features.

Order: Vesle Kjela, Songa, Byrte, Haukeli, Kjela, Lio, Vinje, Tokke, Hogga.

1. **Out-of-fold predictions:** 5 contiguous folds of cases, with 7 cases removed on each side of the validation fold. Overlapping weeks therefore do not leak between train and validation.
2. **Prediction model:** a refit on all training cases predicts the test or Kaggle cases. Each fit averages 3 seeds.
3. **Stacking:** out-of-fold predictions become upstream features in training, and test predictions become upstream features at prediction time. A plant's probability is the mean over its units, and several upstream plants are averaged. The release is that probability × plant capacity, summed over upstream plants.

Upstream predictions flow down the cascade from plant to plant



Topological stacking across the 9 plants

Each arrow adds the 4 commitment features and 2 release features to the plant it points to. The 4 plants on the left have no upstream signals.

Hyperparameters are fixed by hand: 200 trees, depth 6, learning rate 0.05, subsample 0.8, column sample 0.8, histogram tree method.

TCN branch

The network predicts all 14 units for all 168 hours at once, from the features of all 9 plants. It needs no upstream stacking because every plant's features are in its input.

- **Input:** up to 547 channels × 168 hours. Constant channels are removed, then the rest are z-scored and clipped to ±8.

- **Architecture:** 1D dropout on channels, a 1×1 projection to 128 channels, 7 residual blocks, and a 1×1 output to 14 logits per hour.
- **Residual block:** GroupNorm, a non-causal convolution of kernel 3, GELU, dropout 0.2, and a 1×1 convolution. Dilations are 1, 2, 4, 8, 24, 48 and 72.
- **Receptive field:** 319 hours, which covers the whole week from every hour. Non-causal convolutions fit because the week's inputs are known in advance.
- **Scale augmentation:** every training case appears 4 times, with prices and water values multiplied by 1, 0.5, 2 and 4. Features are recomputed for each copy.
- **Training:** binary cross-entropy, AdamW with learning rate 1e-3 and weight decay 1e-2, one-cycle schedule, 10 epochs, batch size 32. Logits are averaged over 3 seeds.

Final output

The final probability comes from a logistic meta-model on the two logits. It has global weights plus small corrections per unit, 44 coefficients and one intercept in total.

```
z_{u,t} = \beta_0 + \beta_u + (w_x + w_{x,u})\,\mathrm{logit}(P^{xgb}_{u,t})
+ (w_c + w_{c,u})\,\mathrm{logit}(P^{tcn}_{u,t}), \qquad \hat{p}_{u,t} =
\sigma(z_{u,t})
```

It is fitted on the local test predictions, 578 cases × 14 units × 168 hours, with C = 1e-4 and no input standardization. The same coefficients are then applied to the Kaggle predictions of both branches.

Combination, local test	Micro AUC
XGBoost alone	0.9825
TCN alone	0.9723
Logit blend, 30% TCN, weight chosen on the test set	0.9851
Logit blend, weight chosen in each training split, 5-fold CV	0.9849
Meta-model, C = 1e-4, 5-fold CV	0.9862
Meta-model, C = 1e-4, trained on first half, scored on second half	0.9869

The recorded Kaggle public score for the meta-model submission is 0.99112. See the first note below about which file it refers to.

Unit	XGBoost	TCN	Blend, 30% TCN
byrte_g1	0.9981	0.9954	0.9983
haukeli_g1	0.9227	0.9279	0.9544
hogga_g1	0.9802	0.9683	0.9806
kjela_g1	0.9799	0.9780	0.9854
lio_g1	0.9421	0.8460	0.9525
songa_g1	0.9803	0.9943	0.9952
tokke_g1	0.9924	0.9840	0.9914
tokke_g2	0.9882	0.9774	0.9878
tokke_g3	0.9918	0.9824	0.9905
tokke_g4	0.9895	0.9800	0.9890
vesle_kjela_g1	0.9946	0.9891	0.9950
vinje_g1	0.9865	0.9750	0.9850
vinje_g2	0.9847	0.9717	0.9829
vinje_g3	0.9796	0.9687	0.9797

The per-unit meta-model AUCs are not saved in the notebook. The two branches are most complementary on Haukeli, Songa and Kjela.